Algebra: Factor & Remainder Theorem

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Name: ______________________________ Date: __________________ Score: / 10

DIRECTIONS *Use the Remainder Theorem to evaluate P(c). Use the Factor Theorem to test factors.*

1 Use Remainder Theorem. Find P(3):

$$P(x) = 3x^2 + 3x + 4$$

Answer: _______________

2 Use Remainder Theorem. Find P(3):

$$P(x) = 3x^2 + 1x - 5$$

Answer: _______________

3 Use Remainder Theorem. Find P(3):

$$P(x) = 3x^2 + 2x + 0$$

Answer: _______________

4 Use Remainder Theorem. Find P(2):

$$P(x) = 2x^2 - 2x - 1$$

Answer: _______________

5 Use Remainder Theorem. Find P(-2):

$$P(x) = 2x^2 - 3x + 0$$

Answer: _______________

6 Is (x–3) a factor?

$$P(x) = x^2 + 2x + 1$$

Answer: _______________

7 Is (x–3) a factor?

$$P(x) = x^2 - 4x + 3$$

Answer: _______________

8 Is (x–3) a factor?

$$P(x) = x^2 + 4x + 3$$

Answer: _______________

9 Is (x–-1) a factor?

$$P(x) = x^2 - 4x + 4$$

Answer:

10 Is (x–2) a factor?

$$P(x) = x^2 - 4$$

Answer:

Answer Key & Solutions

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TEACHER NOTES *P(c) equals the remainder when P(x) is divided by (x–c). If P(c)=0, then (x–c) is a factor.*

1 Use Remainder Theorem. Find P(3):

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$$P(3) = 40$$

$P(x) = 3x^2 + 3x + 4$

2 Use Remainder Theorem. Find P(3):

=

$$P(3) = 25$$

$P(x) = 3x^2 + 1x - 5$

3 Use Remainder Theorem. Find P(3):

=

$$P(3) = 33$$

$P(x) = 3x^2 + 2x + 0$

4 Use Remainder Theorem. Find P(2):

=

$$P(2) = 3$$

$P(x) = 2x^2 - 2x - 1$

5 Use Remainder Theorem. Find P(-2):

=

$$P(-2) = 14$$

$P(x) = 2x^2 - 3x + 0$

6 Is (x–3) a factor?

=

$$P(3) = 16,\ \text{Factor: No}$$

$P(x) = x^2 + 2x + 1$

7 Is (x–3) a factor?

=

$$P(3) = 0,\ \text{Factor: Yes}$$

$P(x) = x^2 - 4x + 3$

8 Is (x–3) a factor?

=

$$P(3) = 24,\ \text{Factor: No}$$

$P(x) = x^2 + 4x + 3$

9 Is (x–-1) a factor?

=

$$P(-1) = 9,\ \text{Factor: No}$$

$P(x) = x^2 - 4x + 4$

10 Is (x–2) a factor?

=

$$P(2) = 0,\ \text{Factor: Yes}$$

$P(x) = x^2 - 4$